FC7xxx PWM Integration Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/07/14	Initial release for MCAL V0.1.0
0.3	2023/09/27	Updated for MCAL V0.3.0
0.4	2023/12/15	Updated for MCAL V0.4.0 - Added new section in memmap and exclusive area in code
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

Table of Contents

Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for the PWM module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Mcu: This module provides the clock reference point for the PWM module.
- Det: This module is necessary for enabling Development error detection.
- EcuC: This module provides configuration for mapping PWM drivers to ECUC partitions (when multi-core functionality is enabled)
- Port: This module provides the port configurations if want output PWM signal through pin.
- Common: This module is the basic module which used to choose the chip.

Module build dependency

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from interrupts (Exclusive Areas).
- Det: This module is necessary for enabling Development error detection.

Module initialization dependency

- Mcu: To use the PMW module, the user needs to initialize the MCU module first
- Port: To output PWM signal, the user needs to initialize PORT module first

2.2 Files Required for Compile

PWM module files:

- _MCAL/Src/Pwm/Src/Pwm.c
- _MCAL/Src/Pwm/Src/Pwm_Ftu.c
- _MCAL/Src/Pwm/Src/Pwm_Hw.c
- _MCAL/Src/Pwm/include/Pwm.h
- _MCAL/Src/Pwm/include/Pwm_Ftu.h
- _MCAL/Src/Pwm/include/Pwm_Ftu_Types.h
- _MCAL/Src/Pwm/include/Pwm_Hw.h
- _MCAL/Src/Pwm/include/Pwm_Hw_Types.h
- _MCAL/Src/Pwm/include/Pwm_MemMap.h
- _MCAL/Src/Pwm/include/Pwm_version.h

- _MCAL_generate/src/Pwm_PBcfg.c
- _MCAL_generate/include/Pwm_PBcfg.h
- _MCAL_generate/include/Pwm_Cfg.h

Common module files:

- _MCAL/Src/Common/include/Aontimer_Reg.h
- _MCAL/Src/Common/include/arm_cortex_asm.h
- _MCAL/Src/Common/include/Common_MemMap.h

- _MCAL/Src/Common/include/Compiler_Cfg.h
- _MCAL/Src/Common/include/CompilerDefinition.h
- _MCAL/Src/Common/include/Cpm_Reg.h
- _MCAL/Src/Common/include/Eth_GeneralTypes.h
- _MCAL/Src/Common/include/Fcpit_Reg.h
- _MCAL/Src/Common/include/Ftu_Common.h
- _MCAL/Src/Common/include/Ftu_Reg.h
- _MCAL/Src/Common/include/Ftu_RegOps.h
- _MCAL/Src/Common/include/Gpio_Reg.h
- _MCAL/Src/Common/include/IRQRouter.h
- _MCAL/Src/Common/include/Mb_Reg.h
- _MCAL/Src/Common/include/Mb_RegOps.h
- _MCAL/Src/Common/include/Mcal.h
- _MCAL/Src/Common/include/Platform_Types.h
- _MCAL/Src/Common/include/Port_Reg.h
- _MCAL/Src/Common/include/Scm_Reg.h
- _MCAL/Src/Common/include/Scm_RegOps.h
- _MCAL/Src/Common/include/SpinLock.h
- _MCAL/Src/Common/include/StdRegMacros.h
- _MCAL/Src/Common/include/Tstmp_Reg.h
- _MCAL/Src/Common/src/Aontimer_Common.c
- _MCAL/Src/Common/src/Ftu_Common.c
- _MCAL/Src/Common/src/IRQRouter.c
- _MCAL/Src/Common/src/Port_Common.c
- _MCAL/Src/Common/src/SpinLock.c

Det module files:

- Det.h

Rte module files:

- SchM_Pwm.h

2.3 Add Plug-ins

PWM module plug-ins are developed for EB tresos Studio, so, to use PWM plug-ins on the EB tresos Studio, the user needs to add the PWM module to the EB plug-ins folder first.

1) Copy the PWM module(_MCAL/EB_Plugins/eclipse/plugins/Pwm) folder to EB tresos plug-ins (EB/tresos/plugins/) folder.
2) Set the PWM module output location folder for the generated source file and header files.
3) Use EB tresos to configure the PWM module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section Name	Section Type	Description
PWM_START_SEC_CONFIG_DATA_8 PWM_STOP_SEC_CONFIG_DATA_8 PWM_START_SEC_CONFIG_DATA_16 PWM_STOP_SEC_CONFIG_DATA_16 PWM_START_SEC_CONFIG_DATA_32 PWM_STOP_SEC_CONFIG_DATA_32	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code(data).
PWM_START_SEC_CONFIG_DATA_UNSPECIFIED PWM_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	Start and stop of Memory Section for Config Data.
PWM_START_SEC_CONST_BOOLEAN PWM_STOP_SEC_CONST_BOOLEAN PWM_START_SEC_CONST_8 PWM_STOP_SEC_CONST_8 PWM_START_SEC_CONST_16 PWM_STOP_SEC_CONST_16 PWM_START_SEC_CONST_32 PWM_STOP_SEC_CONST_32 PWM_START_SEC_CONST_ UNSPECIFIED PWM_STOP_SEC_CONST_ UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only (rodata).
PWM_START_SEC_CODE PWM_STOP_SEC_CODE PWM_START_SEC_ RAMCODE PWM_STOP_SEC_ RAMCODE PWM_START_SEC_CODE_AC PWM_STOP_SEC_CODE_AC	Code	Start and stop of memory Section for Code(text).
PWM_START_SEC_VAR_NO_INIT_BOOLEAN PWM_STOP_SEC_VAR_NO_INIT_BOOLEAN PWM_START_SEC_VAR_NO_INIT_8 PWM_STOP_SEC_VAR_NO_INIT_8 PWM_START_SEC_VAR_NO_INIT_16 PWM_STOP_SEC_VAR_NO_INIT_16 PWM_START_SEC_VAR_NO_INIT_32 PWM_STOP_SEC_VAR_NO_INIT_32 PWM_START_SEC_VAR_NO_INIT_UNSPECIFIED PWM_STOP_SEC_VAR_NO_INIT_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).
PWM_START_SEC_VAR_INIT_BOOLEAN PWM_STOP_SEC_VAR_INIT_BOOLEAN PWM_START_SEC_VAR_INIT_8 PWM_STOP_SEC_VAR_INIT_8	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These

Section Name	Section Type	Description
PWM_START_SEC_VAR_INIT_16 PWM_STOP_SEC_VAR_INIT_16 PWM_START_SEC_VAR_INIT_32 PWM_STOP_SEC_VAR_INIT_32 PWM_START_SEC_VAR_INIT_UNSPECIFIED PWM_STOP_SEC_VAR_INIT_UNSPECIFIED		variables are initialized by startup code (data).
PWM_START_SEC_VAR_NO_INIT_BOOLEAN_NO_CACHEABLE PWM_STOP_SEC_VAR_NO_INIT_BOOLEAN_NO_CACHEABLE PWM_START_SEC_VAR_NO_INIT_8_NO_CACHEABLE PWM_STOP_SEC_VAR_NO_INIT_8_NO_CACHEABLE PWM_START_SEC_VAR_NO_INIT_16_NO_CACHEABLE PWM_STOP_SEC_VAR_NO_INIT_16_NO_CACHEABLE PWM_START_SEC_VAR_NO_INIT_32_NO_CACHEABLE PWM_STOP_SEC_VAR_NO_INIT_32_NO_CACHEABLE PWM_START_SEC_VAR_NO_INIT_UNSPECIFIED_NO_CACHEABLE PWM_STOP_SEC_VAR_NO_INIT_UNSPECIFIED_NO_CACHEABLE	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit and be placed in nocacheable section. These variables are never initialized by startup code (mcal_nocacheable_bss).
PWM_START_SEC_VAR_INIT_BOOLEAN_NO_CACHEABLE PWM_STOP_SEC_VAR_INIT_BOOLEAN_NO_CACHEABLE PWM_START_SEC_VAR_INIT_8_NO_CACHEABLE PWM_STOP_SEC_VAR_INIT_8_NO_CACHEABLE PWM_START_SEC_VAR_INIT_16_NO_CACHEABLE PWM_STOP_SEC_VAR_INIT_16_NO_CACHEABLE PWM_START_SEC_VAR_INIT_32_NO_CACHEABLE PWM_STOP_SEC_VAR_INIT_32_NO_CACHEABLE PWM_START_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE PWM_STOP_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit and be placed in nocacheable section. These variables are initialized by startup code (mcal_nocacheable_data).

Chapter 4 Exclusive Area

PWM module using the services of Schedule Manger (SchM) for entering and exiting critical regions.

The following critical regions are used in the PWM driver:

- Pwm_Hw.c:
  - Pwm_Hw_SetDutyCycle : exclusive area 10
  - Pwm_Hw_SetPeriodAndDuty : exclusive area 11
  - Pwm_Hw_SetOutputToIdle : exclusive area 12
- Pwm_Ftu.c:
  - Pwm_Ftu_SetDutyCycle : exclusive area 0
  - Pwm_Ftu_SetPeriodAndDuty : exclusive area 1
  - Pwm_Ftu_ClearOutputForce : exclusive area 2
  - Pwm_Ftu_SetOutputToIdle : exclusive area 3
  - Pwm_Ftu_DisableNotification : exclusive area 4
  - Pwm_Ftu_EnableNotification : exclusive area 5
  - Pwm_Ftu_EnableTriggerOut : exclusive area 6
  - Pwm_Ftu_DisableTriggerOut : exclusive area 7
  - Pwm_Ftu_MaskOutputs : exclusive area 8
  - Pwm_Ftu_UnMaskOutputs : exclusive area 9
- Pwm.c:
  - Pwm_Init : exclusive area 13
  - Pwm_DeInit : exclusive area 14
  - Pwm_SetDutyCycle : exclusive area 15
  - Pwm_SetPeriodAndDuty : exclusive area 16
  - Pwm_SetOutputToIdle : exclusive area 17
  - Pwm_DisableNotification : exclusive area 18
  - Pwm_EnableNotification : exclusive area 19
  - Pwm_GetChannelState : exclusive area 20
  - Pwm_EnableTriggerOut : exclusive area 21
  - Pwm_DisableTriggerOut : exclusive area 22
  - Pwm_MaskOutputs : exclusive area 23
  - Pwm_UnMaskOutputs : exclusive area 24

- Pwm_SetPowerState : exclusive area 25
- Pwm_GetCurrentPowerState : exclusive area 26
- Pwm_GetTargetPowerState : exclusive area 27
- Pwm_PreparePowerState : exclusive area 28

-

Chapter 5 Interrupt Service Routine (ISR)

Instance	Interrupt Name	IRQ Number (NVIC Interrupt ID)
FTU0	FTU0_IRQHandler	116
FTU1	FTU1_IRQHandler	117
FTU2	FTU2_IRQHandler	118
FTU3	FTU3_IRQHandler	119
FTU4	FTU4_IRQHandler	120
FTU5	FTU5_IRQHandler	121
FTU6	FTU6_IRQHandler	122
FTU7	FTU7_IRQHandler	123
FTU8	FTU8_IRQHandler	124
FTU9	FTU9_IRQHandler	125
FTU10	FTU10_IRQHandler	126
FTU11	FTU11_IRQHandler	127

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
Pwm_Init	PWM_E_ALREADY_INITIALIZED; PWM_E_INIT_FAILED PWM_E_PARAM_CONFIG
Pwm_DeInit	PWM_E_UNINIT
Pwm_SetDutyCycle	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL; PWM_E_DUTYCYCLE_RANGE
Pwm_SetPeriodAndDuty	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL; PWM_E_DUTYCYCLE_RANGE; PWM_E_PERIOD_UNCHANGEABLE; PWM_E_PERIODVALUE
Pwm_SetOutputToIdle	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;
Pwm_GetOutputState	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;
Pwm_DisableNotification	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;
Pwm_EnableNotification	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL; PWM_E_PARAM_NOTIFICATION_NULL; PWM_E_PARAM_NOTIFICATION
Pwm_GetVersionInfo	PWM_E_PARAM_POINTER
Pwm_StartGlobalTime	PWM_E_UNINIT PWM_E_PARAM_GROUP
Pwm_StopGlobalTime	PWM_E_UNINIT PWM_E_PARAM_GROUP
Pwm_GetChannelState	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;
Pwm_EnableTriggerOut	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;
Pwm_DisableTriggerOut	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;
Pwm_MaskOutputs	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;
Pwm_UnMaskOutputs	PWM_E_UNINIT; PWM_E_PARAM_CHANNEL;

Function Name	Error Type
Pwm_SetPowerState	PWM_E_PARAM_POINTER; PWM_E_UNINIT; PWM_E_NOT_DISENGAGED
Pwm_GetCurrentPowerState	PWM_E_PARAM_POINTER; PWM_E_UNINIT;
Pwm_GetTargetPowerState	PWM_E_PARAM_POINTER; PWM_E_UNINIT;
Pwm_PreparePowerState	PWM_E_PARAM_POINTER; PWM_E_UNINIT;

6.2 Dem

None

Chapter 7 Function Calls

7.1 Function Calls during Startup

The API needs to be called is PWM_Init(NULL_PTR);

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

None

Chapter 8 Other Requirements

8.1 Notification, Callback, Callout

- Notification

  If the user configures notifications and the value is not "NULL_PTR" or "NULL", an extern declaration will generate in Pwm_Cfg.h. User need implement the notification in any file which include Pwm_Cfg.h.

- Callback

  None

- Callout

  None

8.2 Macros

Please check various definitions available in Common module's include file Mcal.h for details.

- If OS is not used:

  AUTOSAR_OS_NOT_USED needs to be defined.

  In this case, If USE_SW_VECTOR_MODE is not defined, the user needs to map the interrupt vector table function to ISR function, for example：

```
extern ISR(FTU_0_ISR);

void FTU0_IRQHandler(void)

{

FTU_0_ISR();

}
```

- If OS is used:

  Do not define AUTOSAR_OS_NOT_USED.

Chapter 9 Integration Steps

1) Configure the PWM module and generate configuration files (please refer to Building chapter for details).
2) Configure appropriate memory sections in linker file or other (please refer to Memory chapter for details).
3) Map interrupt notification to their vector locations (please refer to chapter ISR for details).
4) Build the PWM module with dependent modules.